

Unit 2, Lesson 1: Finding the Greatest Common Factor

Benchmark

MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.

Learning Targets

- ☐ I can find the greatest common factor (GCF) in mathematical and real-world contexts.
- ☐ I can rewrite fractions in simplest form using the greatest common factor (GCF) of the numerator and denominator.

2.1.1 Warm-Up: Where Do Math Symbols Come From?

Math is full of symbols that include lines, dots, arrows, English letters, Greek letters, and more. It can look and sometimes feel like its own language, and in many ways it is!

Robert Recorde invented the equals sign, $=$, because he was tired of writing “is equal to” over and over again.

1. What is a word or phrase that you use a symbol to represent instead of writing it out each time?
2. Why did early mathematicians decide to use symbols to express operations, phrases, and mathematical ideas?
3. Why are math symbols still helpful to us today?



2.1.2 Guided Instruction: Greatest Common Factor

1. Marcus and Tonya are making care packages for soldiers overseas. Inside of each package, they will include popular snacks and handwritten letters. They have 16 snacks and 12 letters to put in the care packages. They want each care package to be equally sized, containing both snacks and letters, with nothing left over.
- a. Complete the tables below to show the possible combinations when making each care package.

Distributing Snacks Equally	
Number of Care Packages	Number of Snacks per Package
1	16
2	8
4	
	1

Distributing Letters Equally	
Number of Care Packages	Number of Letters per Package
1	12
2	6
3	
	3
	2
	1

- b. What is the greatest number of care packages that can be created using **only** snacks?

- c. What is the greatest number of care packages that can be created using **only** letters?
- d. What is the greatest number of care packages that can be created using **both** snacks and letters?



A **factor** is a number that divides into another number exactly and without leaving a remainder.

For example, the **factors** of 20 are 1, 2, 4, 5, 10, and 20.

The numbers (or expressions) that are being multiplied are called the factors. When numbers (or expressions) are multiplied, the answer is called a product.



factor $\times$ **factor** = **product**

- e. How were the factors of 12 and 16 used in the previous problem?
- The factors of 16 were used to determine the whole number of

- ☐ letters
 - ☐ snacks

 that could be included, depending on the number of care packages.

- The factors of 12 were used to determine the whole number of

○ letters

○ snacks

 that could be included, depending on the number of care packages.



The **greatest common factor**, or **GCF**, is the largest common factor of two or more numbers.

An example of how to find the **greatest common factor (GCF)** is shown.

The factors of 15 are 1, 3, 5, and 15.

The factors of 20 are 1, 2, 4, 5, 10, and 20.

The GCF of 15 and 20 is 5.

- f. Complete the statement to explain what the GCF of 16 and 12 is from the previous problem.

The GCF of 16 and 12 is _____, which represents the greatest number of

care packages that can be created using

○ only snacks.

○ only letters.

○ both snacks and letters.

2. Find the greatest common factor of 6 and 18.

	Factors	GCF:
6		
18		

2.1.3 Try It: Greatest Common Factor

1. Find the greatest common factor of each pair of numbers.
 - a. 15 and 200
 - b. 123 and 41
 - c. 54 and 36
 - d. 282 and 108

2. Penelope is organizing the teacher-student relay race. The teams must be divided evenly, with no participants left out. If there are 56 students registered and 21 teachers, what is the greatest number of teacher-student teams that will be able to participate in the relay race?

2.1.4 Guided Instruction: Writing Equivalent Fractions Using GCF

The GCF can also be used to rewrite fractions. When using the GCF to write an equivalent fraction, the result is often said to be in “simplest” form.



To **simplify a fraction**, generate an equivalent fraction by dividing the numerator and denominator by the **GCF** of the numerator and denominator.

1. In the fraction $\frac{25}{90}$, the numerator is 25 and the denominator is 90.
 - a. What is the greatest common factor of 25 and 90?
 - b. Divide the numerator and denominator by the GCF. What is the resulting fraction?
 - c. Complete the statement to describe the relationship between the fractions.

After dividing the numerator and denominator of $\frac{25}{90}$ by their GCF, the resulting fraction,

$\frac{\quad}{\quad}$

,

☐ is
☐ is not

 equivalent to $\frac{25}{90}$, rewritten in simplest form.

2.1.5 Try It: Writing Equivalent Fractions using GCF

1. Rewrite each fraction in simplest form using the GCF of the numerator and denominator.

a. $\frac{8}{28}$

b. $\frac{45}{90}$

c. $\frac{16}{34}$

2. Mikayla states that $\frac{900}{1000}$ is not the same value as $\frac{9}{10}$, because they are fractions of different sizes. Is she correct? Provide an explanation to support your answer.

2.1.6 Your Turn!

1. Find the greatest common factor (GCF) of each pair of numbers.
 - a. 18 and 24
 - b. 45 and 60
 - c. 665 and 30

2. Rewrite each fraction in simplest form using the GCF of the numerator and denominator.
 - a. $\frac{29}{87}$
 - b. $\frac{10}{15}$
 - c. $\frac{175}{210}$

3. Mrs. Chaunuan is making gift bags filled with pencils and stickers for class prizes. Mrs. Chaunuan has 24 pencils and 36 stickers to use. Each bag will have the same number of each item, with no items left over. What is the greatest number of gift bags she can make?

2.1.7 Wrap-Up: Ticket Out the Door

1. Rewrite $\frac{52}{104}$ in simplest form.

2. Diego is preparing brownies and cookies for a bake sale. He has 48 brownies and 64 cookies. Diego wants to make bake sale bags. Each bag will have the same amount of each item, with no items left over. What is the greatest number of bake sale bags he can make?